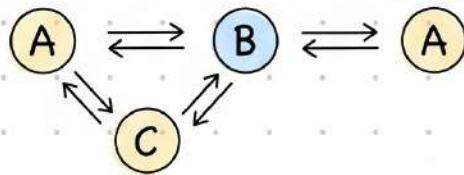


Thermodynamics: Fundamental Laws

Zeroth Law:

Zeroth law of thermodynamics states that two systems in thermal equilibrium with a third system are in thermal equilibrium with each other. Zeroth law leads to the idea of temperature.



First Law:

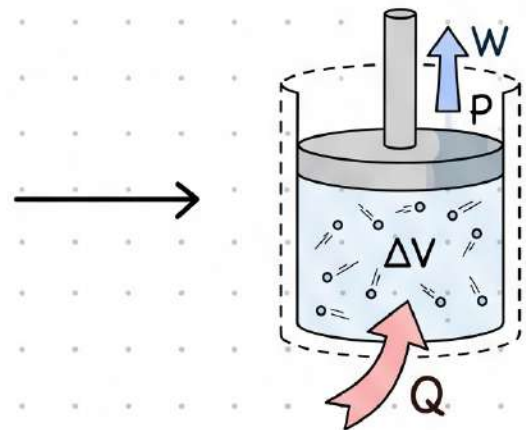
First law of thermodynamics states that when heat Q is added to a system while the system does work W , the internal energy U changes by an amount equal to $Q - W$.

First law of thermodynamics is general law of conservation of energy.

$$\Delta Q = \Delta U + \Delta W$$

Since Work done $\Delta W = P\Delta V$, the equation becomes

$$\Delta Q = \Delta U + P\Delta V.$$



Polytropic Process:

It is a thermodynamic process that can be expressed as $PV^x = \text{Constant}$, where x is the polytropic exponent.

Work done in a general polytropic process: $\left[\frac{nR\Delta T}{1-x} \right]$

Molar heat capacity for the process: $C = \frac{f}{2}R + \frac{R}{1-x}$

Standard Processes: Isobaric & Isochoric

Isobaric Process (Constant Pressure)

Conditions: $dP = 0$, Equation of State:

$$\frac{V}{T} = c$$

Work Done (W):

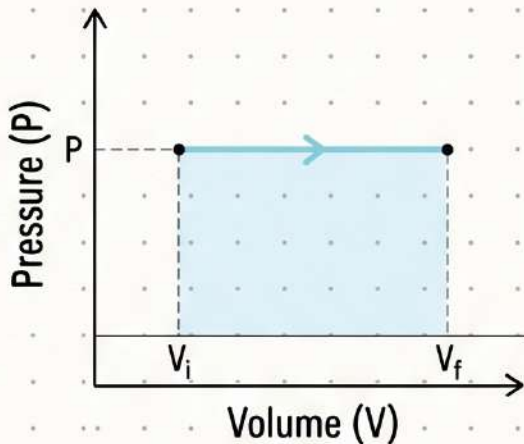
$$P(V_f - V_i) = nR(T_f - T_i)$$

Change in Internal Energy (ΔU):

$$\frac{f}{2} nR(T_f - T_i) = \frac{f}{2} P(V_f - V_i)$$

Heat Exchanged (ΔQ):

$$\left(\frac{f}{2} + 1\right)nR\Delta T = \left(\frac{f}{2} + 1\right)P(V_f - V_i)$$



Isochoric Process (Constant Volume)

Conditions: $dV = 0$, Equation of State:

$$\frac{P}{T} = c$$

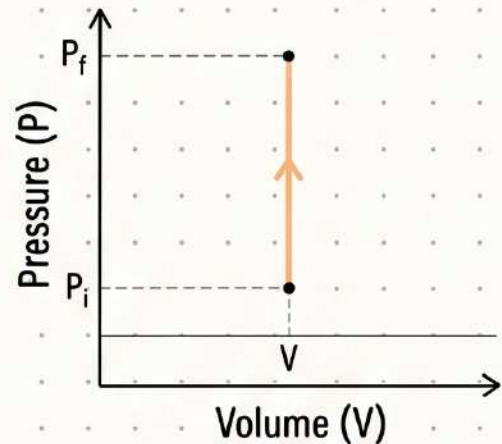
Work Done (W): 0

Change in Internal Energy (ΔU):

$$\frac{f}{2} nR(T_f - T_i) = \frac{f}{2} V(P_f - P_i)$$

Heat Exchanged (ΔQ):

$$\frac{f}{2} nR(T_f - T_i) = \frac{f}{2} V(P_f - P_i)$$



Standard Processes: Isothermal & Adiabatic

Isothermal Process (Constant Temperature)

Conditions: $dT = 0$

Equation of State: $PV = c$

Work Done (W):

$$W = nRT \ln\left(\frac{V_f}{V_i}\right) = nRT \ln\left(\frac{P_i}{P_f}\right)$$

Change in Internal Energy (ΔU):

$\Delta U = 0$ (For isothermal process $\Delta T = 0$)

Heat Exchanged (ΔQ):

$$\Delta Q = W = nRT \ln\left(\frac{V_f}{V_i}\right) = nRT \ln\left(\frac{P_i}{P_f}\right)$$

Adiabatic Process (Zero Heat Transfer)

Conditions: $dQ = 0$

(For adiabatic process $\Delta Q = 0$)

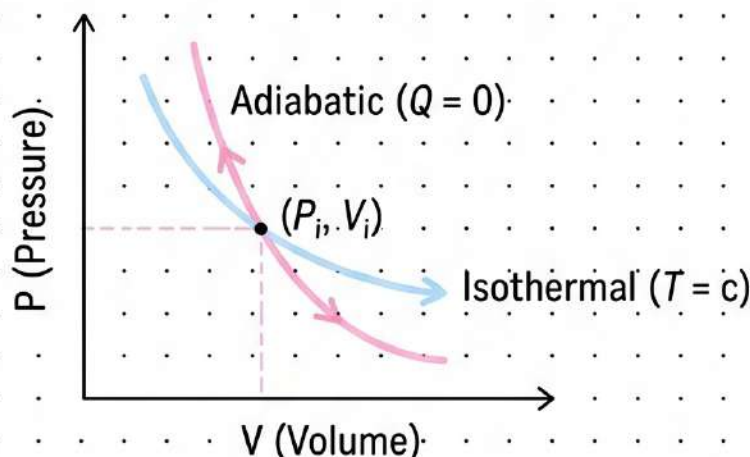
Equation of State: $PV^\gamma = c$

Work Done (W):

$$W = \frac{f}{2} nR(T_i - T_f) = \frac{f}{2} (P_i V_i - P_f V_f)$$

Change in Internal Energy (ΔU):

$$\Delta U = \frac{f}{2} nR(T_f - T_i)$$



Slope Comparison: Slope of adiabatic = $\gamma \times$ (slope of isotherm)

2nd Law, Engines & Refrigerators

Second Law of Thermodynamics:

Clausius statement: No process is possible whose only result is the transfer of heat from a colder object to a hotter object.

Kelvin-Planck statement: No process is possible whose only result is the absorption of heat from a reservoir and complete conversion of the heat into work. No engine can have efficiency equal to 1 or no refrigerator can have co-efficient of performance equal to infinity.

Carnot Engine (Ideal Engine):

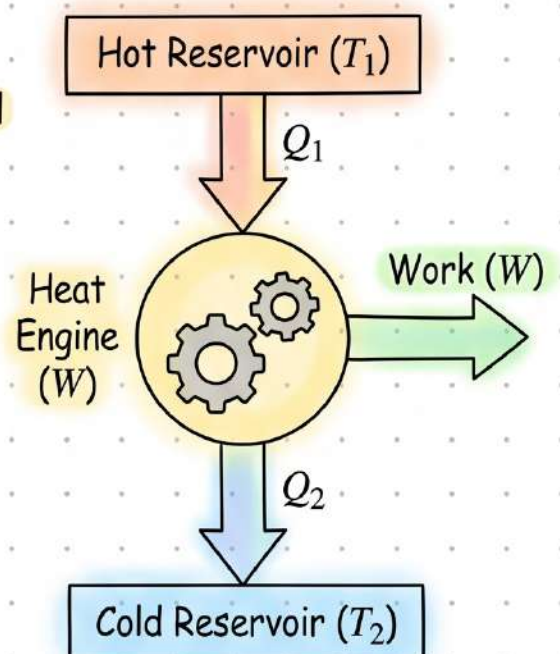
The Carnot cycle consists of two reversible isothermal process and two reversible adiabatic process.

Work (W): $W = Q_1 - Q_2$

Efficiency (η): $\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$

$$\eta = \frac{W}{Q_1}$$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$



Refrigerator & Heat Pump:

Refrigerator Coefficient of performance (β):

$$\beta = \frac{Q_2}{Q_1 - Q_2} = \frac{Q_2}{W}$$

$$\beta = \frac{1 - \eta}{\eta}$$

Heat pump (α):

$$\alpha = \frac{Q_1}{W} = \frac{Q_1}{Q_1 - Q_2} = \frac{1}{\eta}$$

Sign Conventions:

If $Q > 0$, heat is added to the system;

If $Q < 0$, heat is removed from the system.

If $W > 0$, work is done by the system (Expansion);

If $W < 0$, work is done on the system (Compression).